

6.104 · Software Design · MIT · Fall 2026

# how to innovate

Daniel Jackson

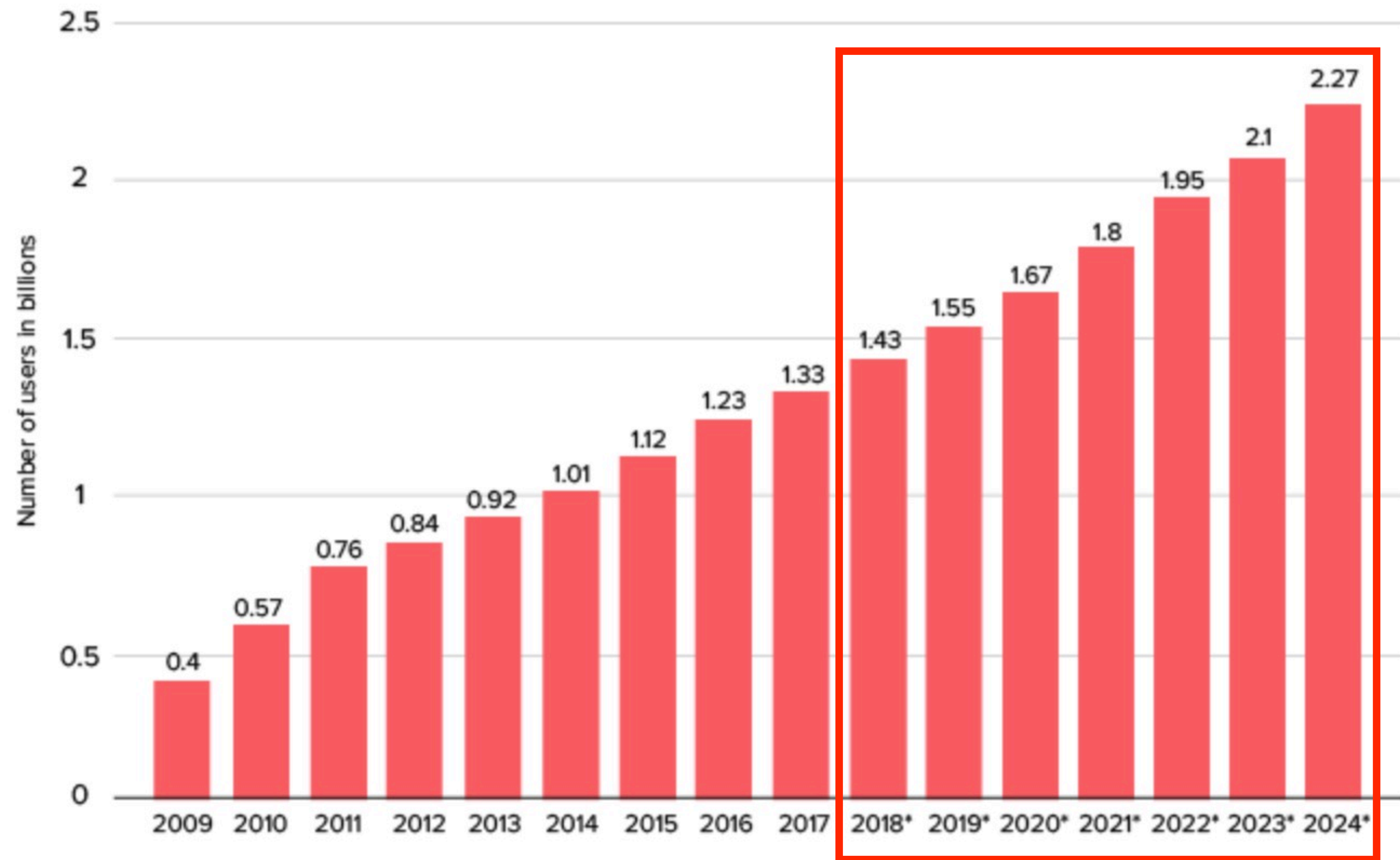
# the key to successful innovation

today you'll learn the **one thing**  
to do **when you start** a project  
that maximizes the **chance of success**

*what success  
looks like*



Number of estimated Skype users registered worldwide from 2009 to 2024 (in billions)

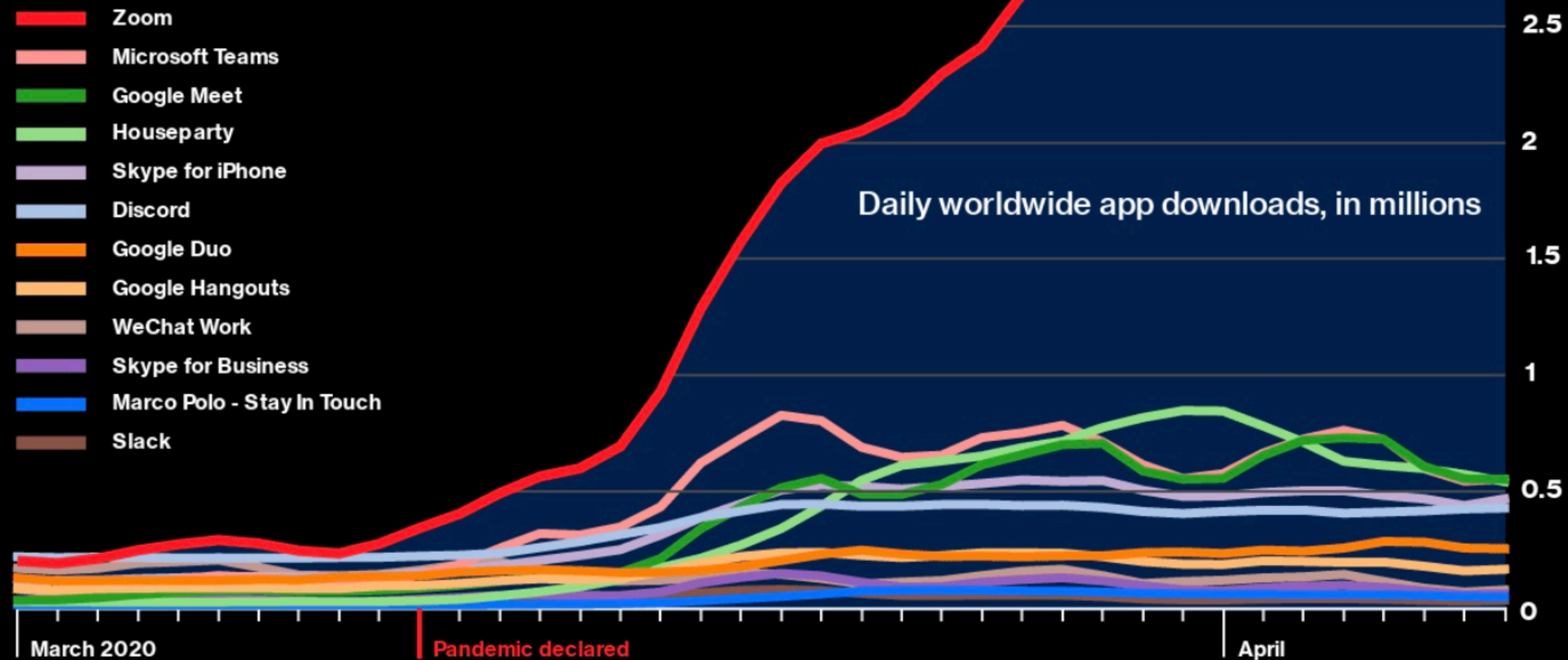


from Statistica: 2018-2024 estimated

# Covid-19 / Zoom-a-Zoom-Zoom

BBOViz  
Covid-19

There's only one winner in the  
work-from-home meeting app wars



Source: Apptopia

BEN SCHOTT

BloombergOpinion

*want or need?*



# YC's motto (paul graham, 2005)



poster designed by Rudy Adler

see: Paul Graham, Be Good (<https://paulgraham.com/good.html>)

# alan kay visits YC (2015)



**Sam Altman** ✓



@sama

Alan Kay visits YC office, grafittis [@paulg](#)'s poster





*a fool's errand?*

We speak of needs as though they exist in some real or physical way. We say “I need a car,” or “I need a house.” In actual fact, cars and houses are not needs in themselves, they are but one way to meet certain needs such as mobility or shelter. The need itself is a perceived lack, something that is missing. Needfinding is thus a paradoxical activity—what is sought is a circumstance where something is missing. In order to find and articulate a need, this missing thing must be seen and recognized by someone.

Rolf Faste, Perceiving Needs (1987)

*a better take*



FOREWORD BY ARVIND KRISHNA, CHAIRMAN AND CEO, IBM

# THE HEART OF INNOVATION

A Field Guide for Navigating to  
Authentic Demand



MATT CHANOFF, MERRICK FURST,  
DANIEL SABBAAH, AND MARK WEGMAN

## the “waking dream”

you know who your customers are & how they think  
you see people buying and using your product  
but it’s a waking dream: the real world is different

## how does this happen?

“being wrong feels like being right”  
your thinking is skewed by cognitive biases

## so what to do?

recognize this reality: you can’t fix it, but you can work with it  
look for indifference, not confirmation  
find a “not not”: not OK for people to not do something

## where do you start?

examine deeply the situation your user is in

# so what exactly is wrong with needs & wants?

## **they're invisible**

you're looking for something that doesn't exist

## **we are delusional**

we say we want and even need things, but then...

## **solutionism**

needs & wants are often solutions in disguise



# starting from bad situations instead

## **what is a bad situation?**

a concrete scenario that a person finds themselves in that makes them uncomfortable, upset, irritated, etc because they are forced to behave in a certain way

## **so what do people do?**

tolerate the bad situation  
find a workaround  
avoid the situation

## **what is innovation?**

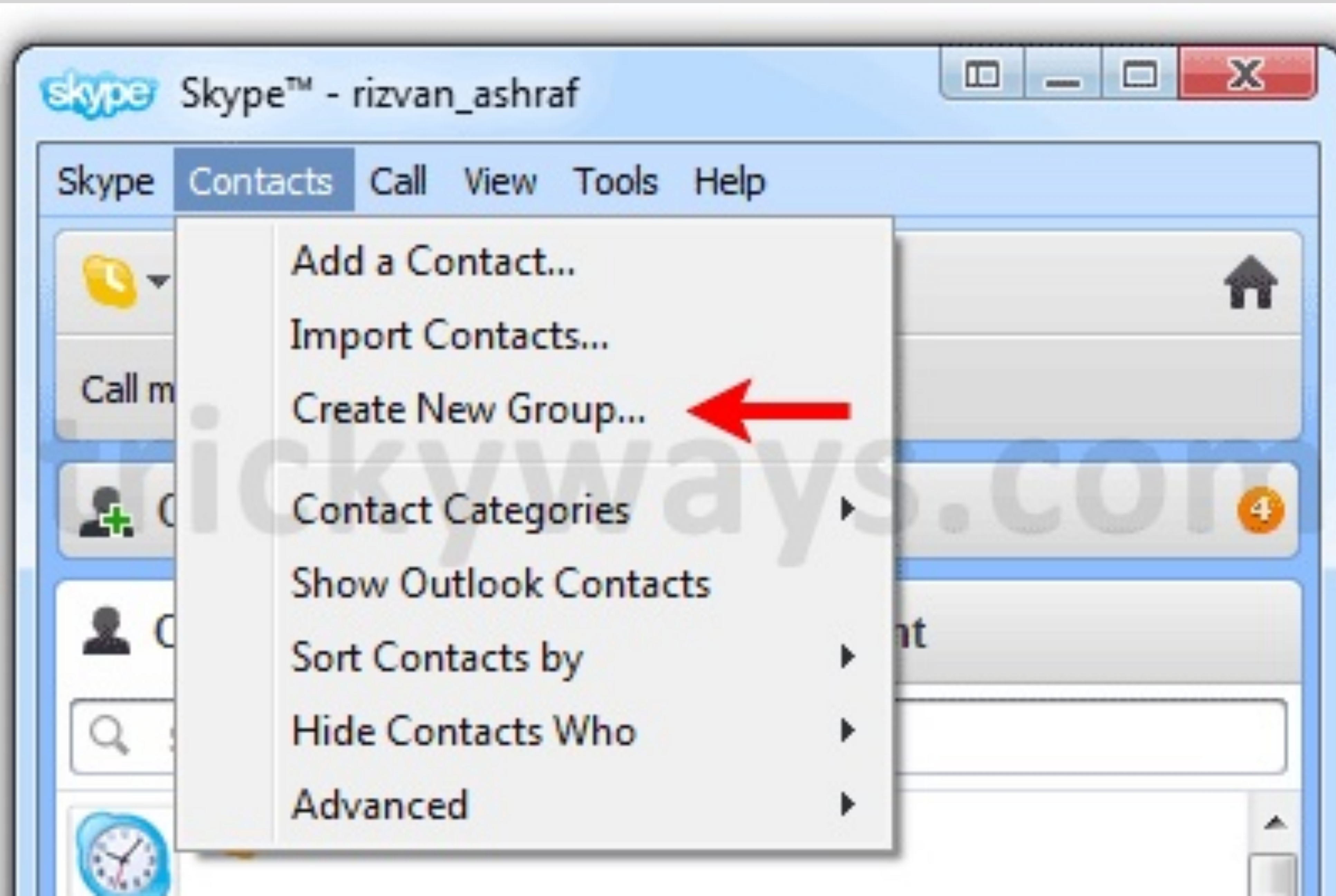
preventing or mitigating bad situations  
letting people behave the way they always wanted to  
(even if they didn't know it)

*bad situations*

*an app*

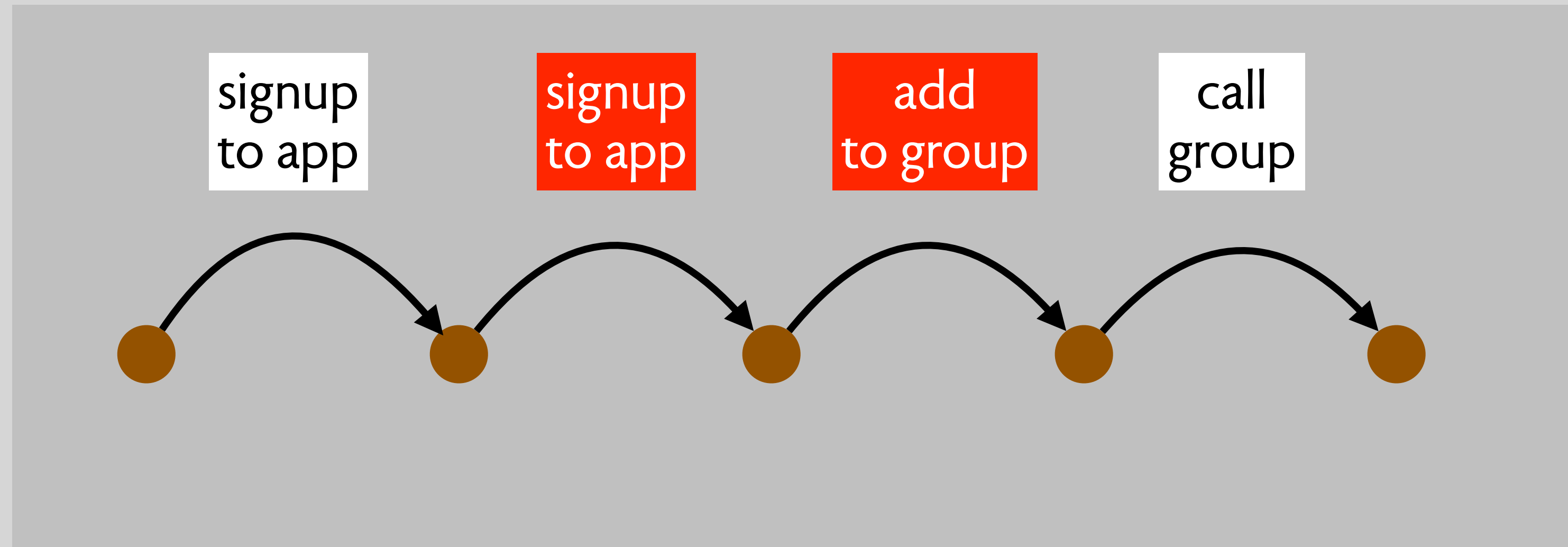


what was zoom's bad situation?

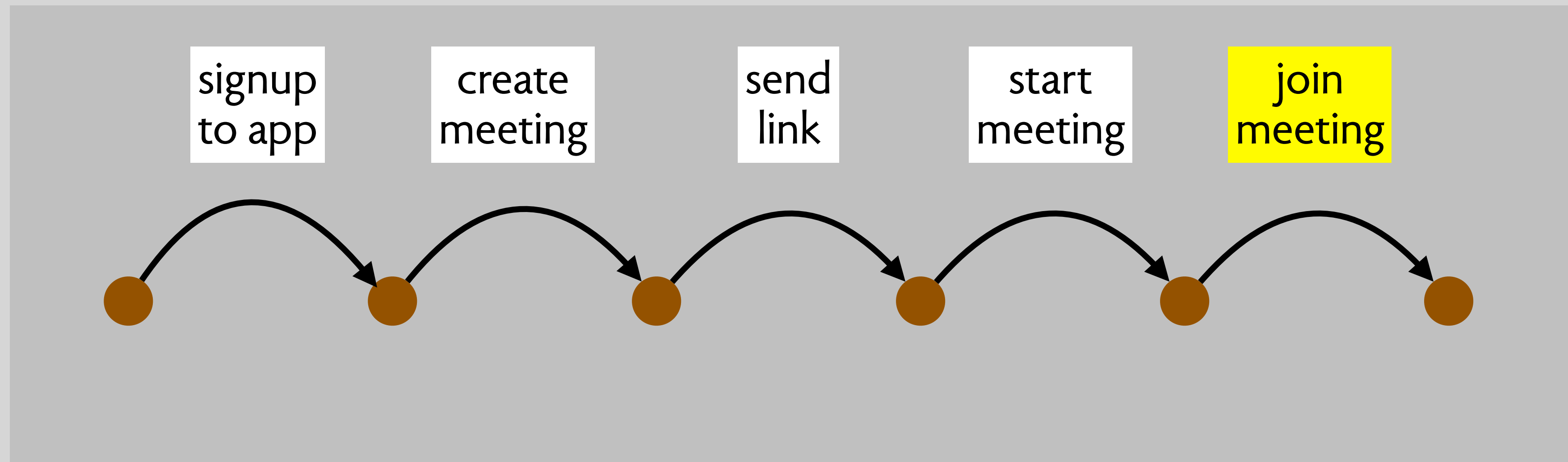


# zoom's fix: self joining

before



after





a bit too easy to join...



March 31, 2020

*a device*

# what made the iPod great?



**Apple iPod**  
(Jony Ive, 2001)



**Braun Pocket Radio T3**  
(Dieter Rams, 1958)





AI image from nano banana

the bad situation: can't have all your music with you





a timely technology



iPod 5GB drive  
(Toshiba, 2001)



FireWire serial bus  
(Apple, 1995)

a gadget





bad #1

bad #2

*an app feature*

what bad situation(s) does this solve?



## Sign in to GitHub

**Username or email address**

**Password**

[Forgot password?](#)

**Sign in**



*activity*

*when fixes cause new bad situations*

# bad situation #1



what is the bad situation that motivates file backup?

## bad situation #2

**how to keep all the photos you take**  
upload your photos to *import/*  
select photos to keep and move to *photos/*  
run backup, then delete *import/*\*



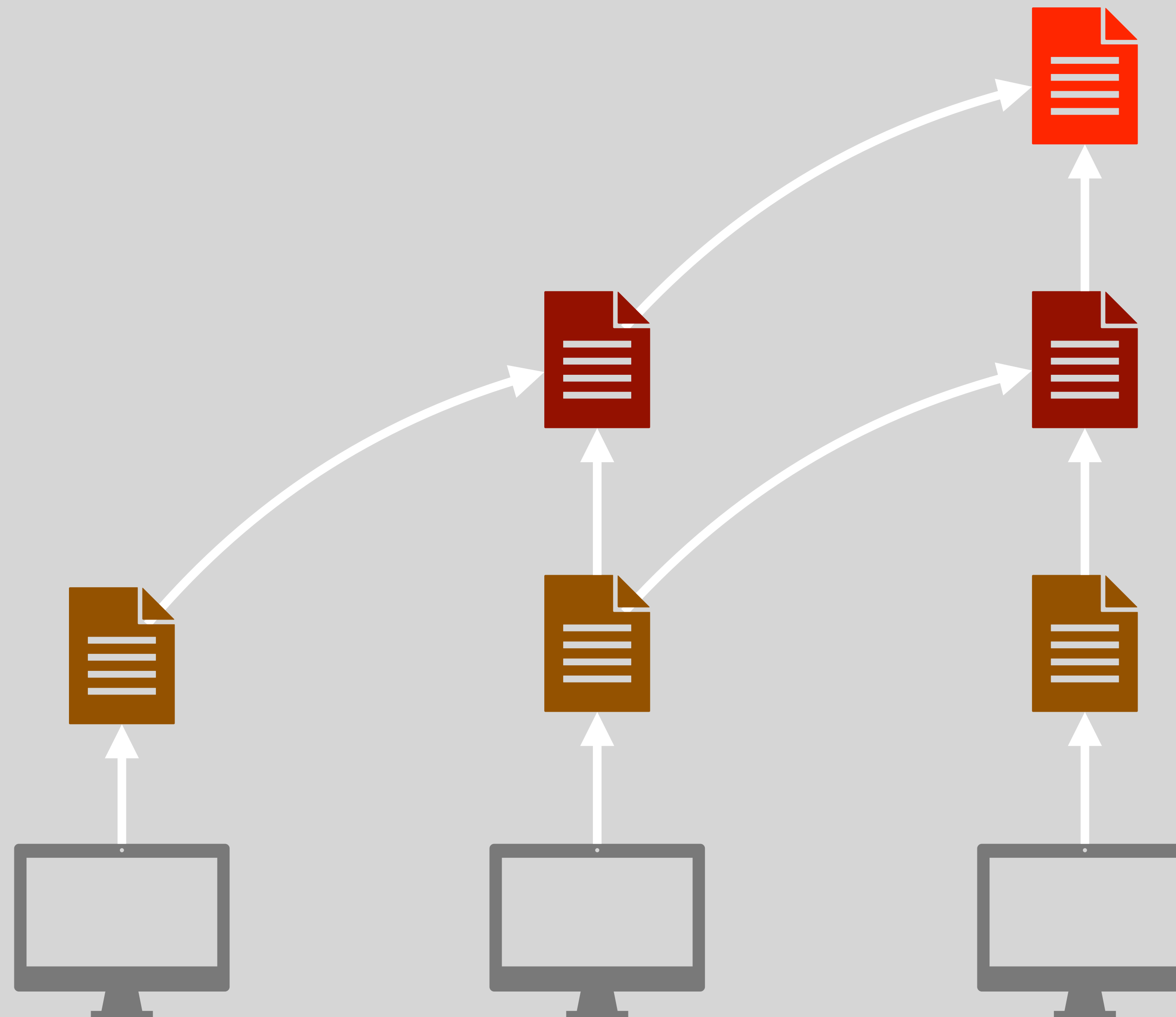
fix: the backup service  
removes old versions or deleted files  
from backup after 30 days



**bad situation #3**



# sketching a solution



**diff of diff**  
atypical changes

**checksum diff**  
files added, deleted, changed

**checksum**  
fingerprint of each file

common pitfalls

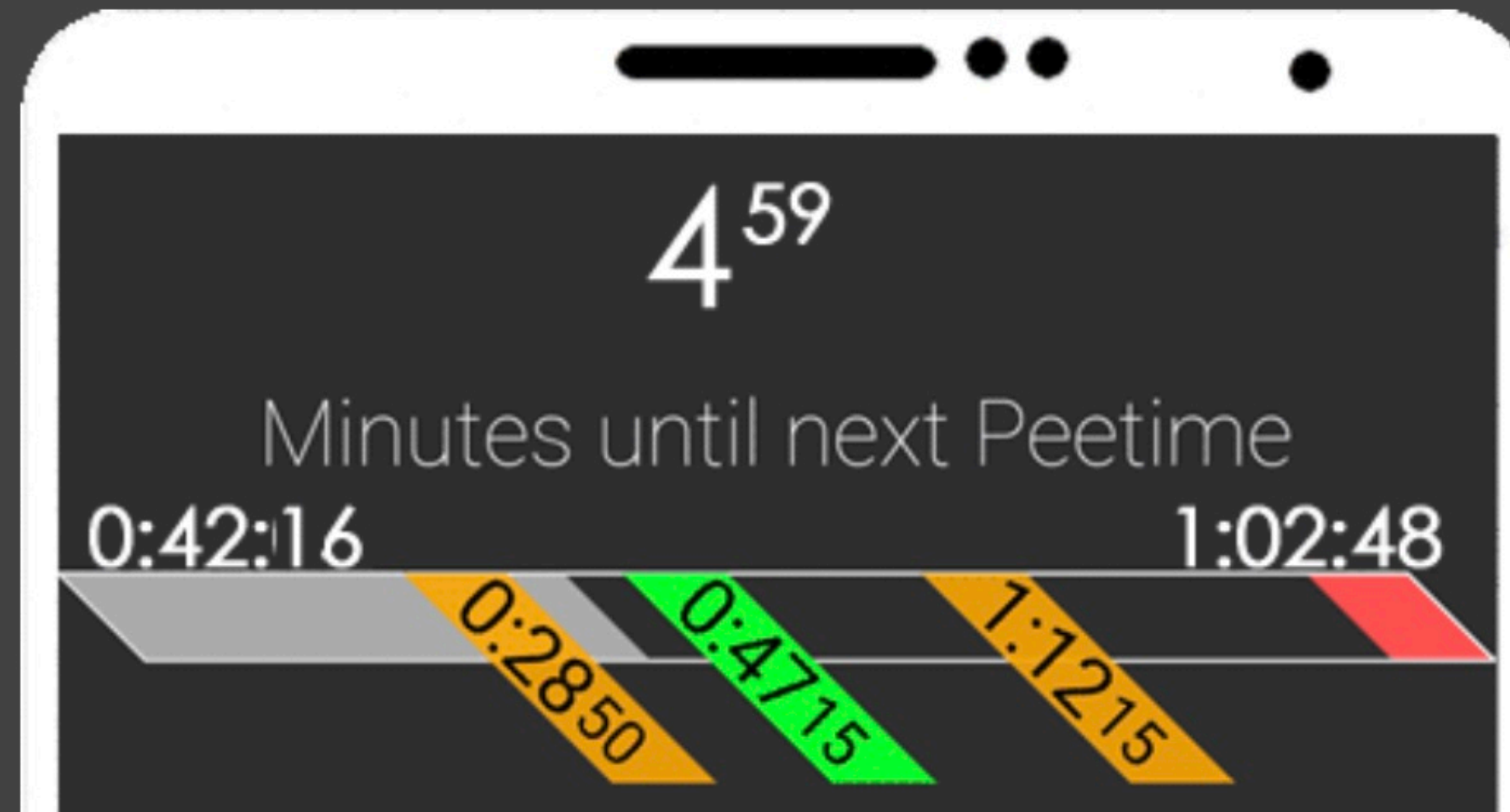
bad, but not  
bad enough



# The World's Most Indispensable Movie App

The RunPee app tells you the best times to  
**run & pee during a movie**  
so you don't miss the best scenes.

As seen on





# Carrier Pidge sends messages at pigeon speed.

Carrier Pidge is a slower social messaging experience for notes that deserve a route, a little suspense, and a moment of arrival.

[Release your first pigeon →](#)[How it works](#)

## A message can be more than a notification.

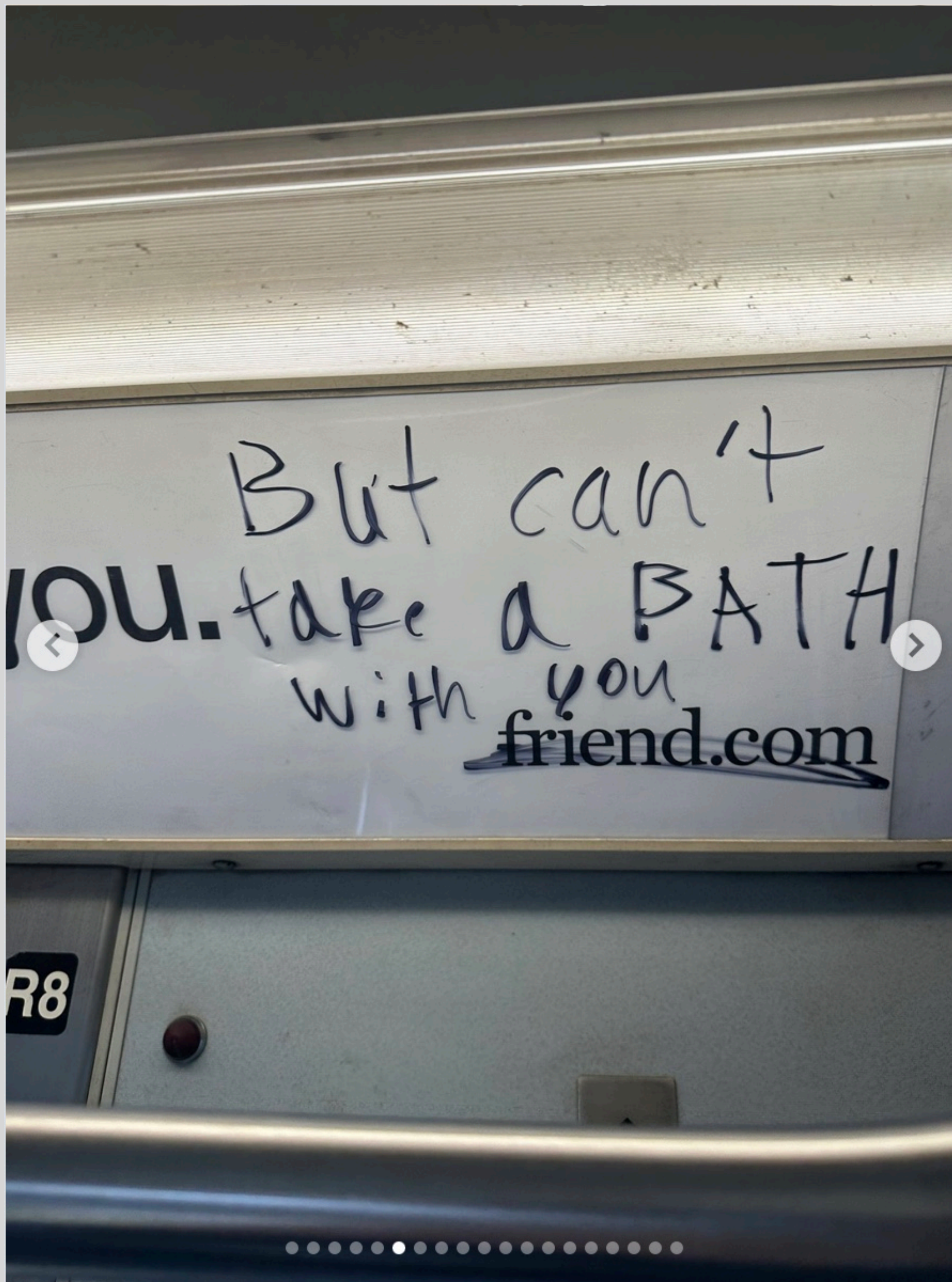
The web is full of instant replies. Carrier Pidge makes room for the other feeling: writing something personal, releasing it, and letting the distance become part of the story.

*antibodies*



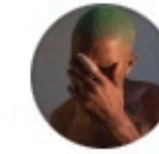






aplasticplant

Follow ...



aplasticplant 49w

the largest nyc subway advertising campaign ever is for a fake ai friend you wear around your neck to keep you company, and people have been covering the ads with graffiti.

avi schiffmann, ceo of the ai companion startup, poured more than \$1 million dollars into a new york city subway takeover plastering stations and cars across all five boroughs.

friend.com is a startup that's developing an ai-powered wearable pendant designed to act as a constant companion. the device, a small disc worn around the neck, connects to an iphone app and uses large language models like google's gemini to listen, comment, and engage in conversation throughout the day.

92.3K 871 3.3K

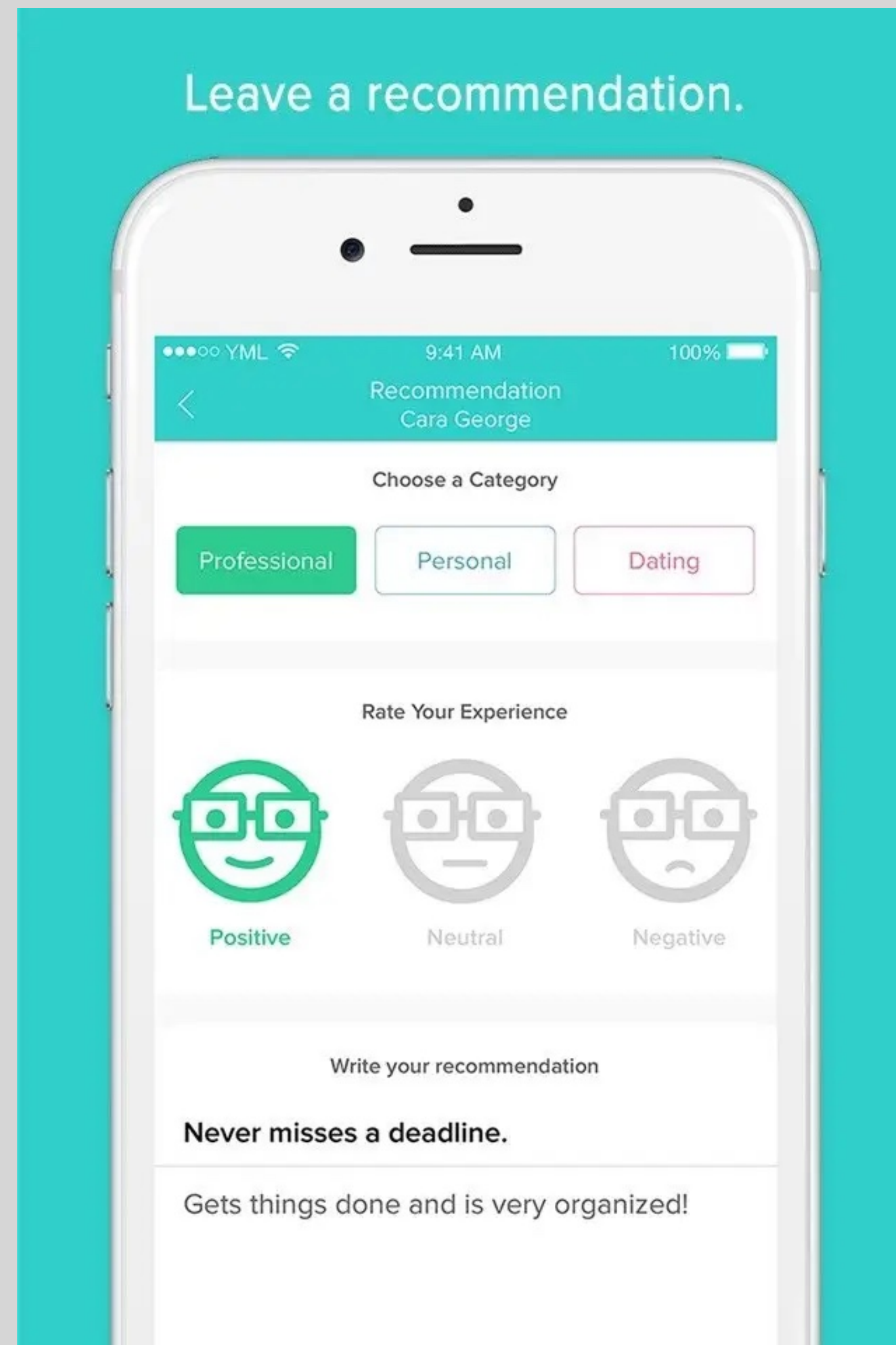
September 27, 2025



Add a comment...







Peeples: "Yelp for people" (2015)

antibodies activated in other stakeholders

TECHNOLOGY

# Google: How not to be a 'Glasshole'

by: [CNN Wire](#)

Posted: Feb 19, 2014 / 03:18 PM MST

Updated: Feb 19, 2014 / 03:18 PM MST



*(Photo: Courtesy Google)*

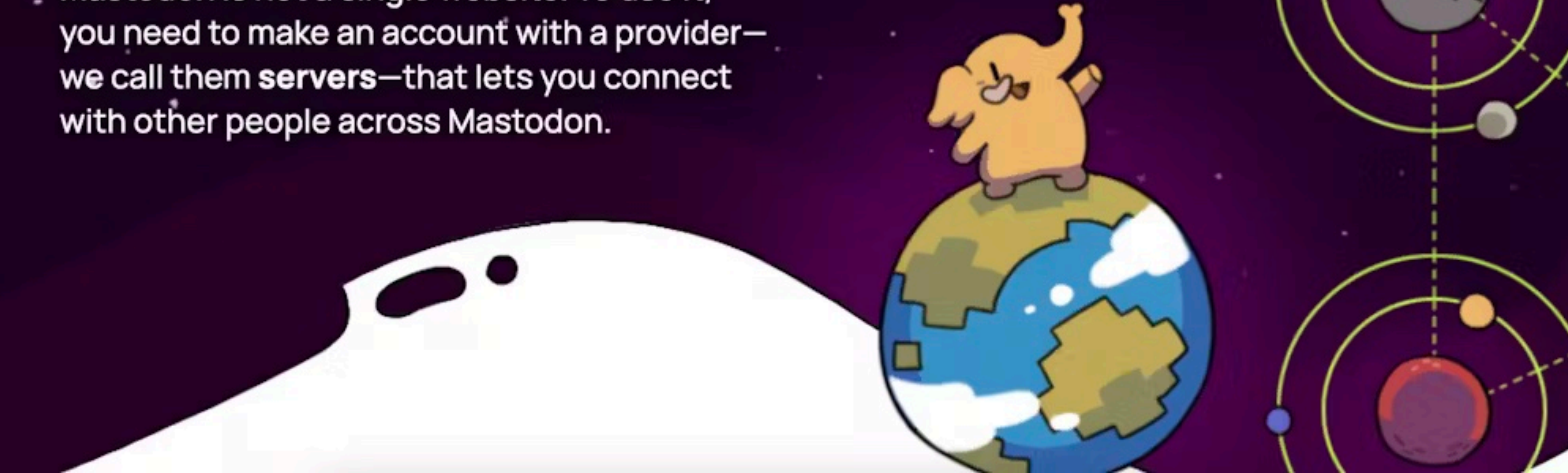
**Google Glass (2014-2015)**  
negative reaction from bystanders  
being recorded felt creepy



*friction*

# Servers

Mastodon is not a single website. To use it, you need to make an account with a provider—we call them **servers**—that lets you connect with other people across Mastodon.



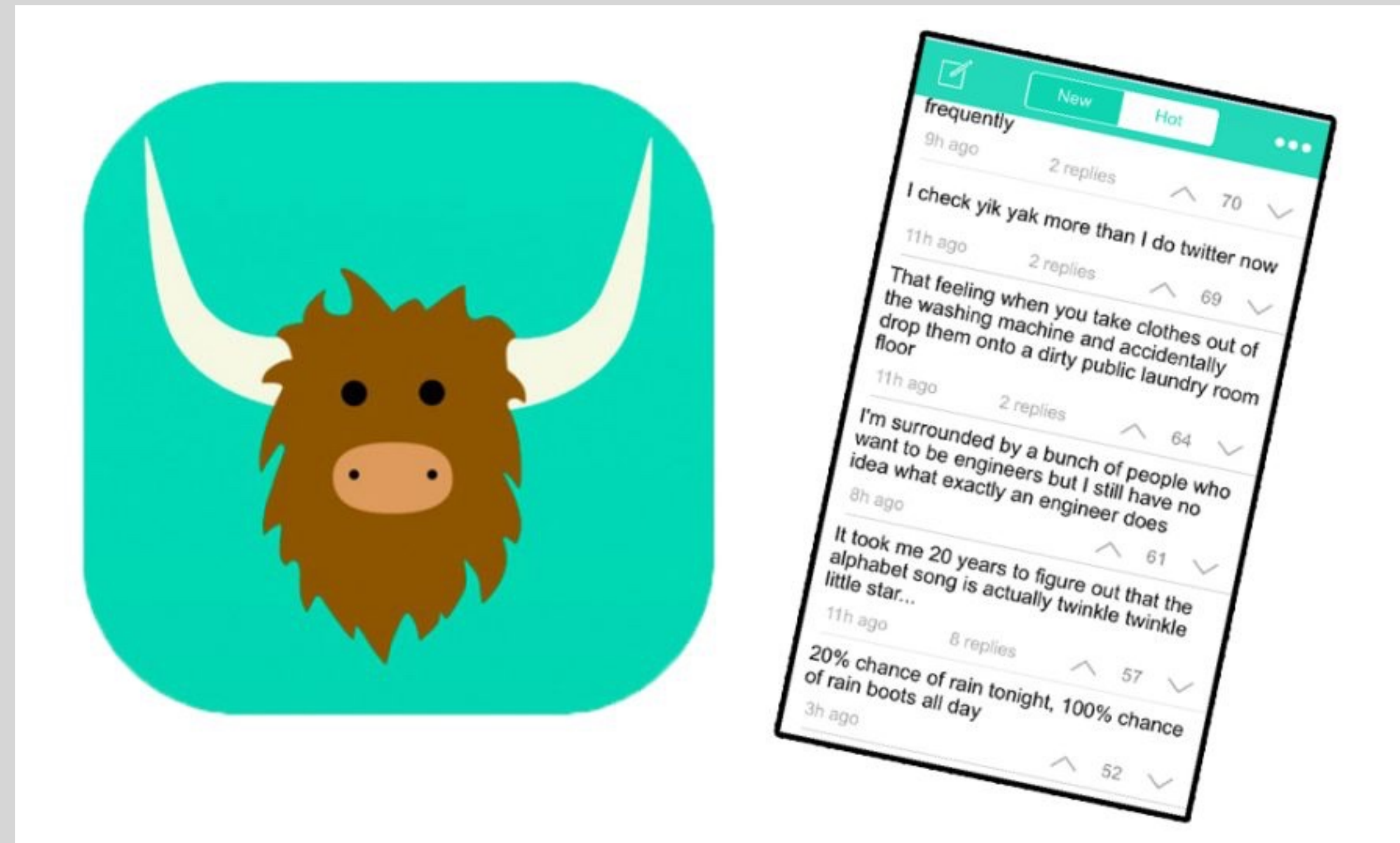
**Mastodon** (since 2016)  
open source social network  
“federated”: servers managed independently

selecting a server too hard for most  
peaked at >2m in 2022, but now <1m

unintended  
consequences



# unexpected consequences



## **YikYak** (2013-2017)

location-based, anonymous social media  
“yaks” visible to others within 5 mile radius  
shutdown due to cyberbullying and hate speech

*your assignment*

# what you'll do in your assignment



## **explore your surroundings**

clubs, activities, services  
hobbies, special interests



## **notice bad situations**

what's not working?  
what's annoying?

maybe other people  
**haven't noticed**



## **collect evidence**

data, complaints, comps  
seek disconfirmation



## **sketch solution**

plausible & fixes bad situation  
novel in *some* respect



parting  
thought

“To design something really well, you have to get it. You have to really grok what it’s all about. It takes a passionate commitment to really thoroughly understand something, chew it up, not just quickly swallow it. Most people don’t take the time to do that.”

Steve Jobs